

1. Between collisions, the oscillator has amplitude decreasing as  $e^{-\Gamma t/2}$  at frequency  $\omega_0$ . When collision happens, a random phase  $\phi(t)$  is added to the oscillator, i.e.,

$$x(t) = x_0 e^{-\Gamma t/2} e^{-i\omega_0 t} e^{i\phi(t)} \quad t > 0 \quad (1)$$

where  $\phi(t)$  is constant between collisions and changes by a random amount at each collision.

The Fourier amplitude is then

$$x(\omega) = \int_0^\infty x(t) e^{i\omega t} dt \quad (2)$$

and the energy spectrum is

$$\frac{dI}{d\omega} \propto \langle |x(\omega)|^2 \rangle = \int_0^\infty \int_0^\infty x_0^2 e^{-\Gamma(t+t')/2} e^{i(\omega-\omega_0)(t'-t)} \langle e^{i[\phi(t')-\phi(t)]} \rangle dt dt' \quad (3)$$

From the given distribution, the probability that no collision occurs within an interval of length  $|t' - t|$  (the next collision lands beyond it) is

$$P_0(|t' - t|) = \int_{|t'-t|}^\infty \nu e^{-\nu T} dT = e^{-\nu|t'-t|} \quad (4)$$

If there is at least one collision in the interval, the phase difference from each collision is a random number  $\Delta\phi$  between 0 and  $2\pi$  so the mean of  $e^{i\Delta\phi}$  is zero. Therefore only when no collision occurs in the interval does the phase difference contribute to the average, i.e.,

$$\langle e^{i[\phi(t')-\phi(t)]} \rangle = e^{-\nu|t'-t|} \cdot 1 + (1 - e^{-\nu|t'-t|}) \cdot 0 = e^{-\nu|t'-t|} \quad (5)$$

Thus

$$\langle |x(\omega)|^2 \rangle = x_0^2 \int_0^\infty \int_0^\infty e^{-\Gamma(t+t')/2} e^{i(\omega-\omega_0)(t'-t)} e^{-\nu|t'-t|} dt dt' \quad (6)$$

Exchanging  $t$  and  $t'$  makes the integrand complex conjugate, so with  $t' = t + s, s > 0$ , we have

$$\begin{aligned} \langle |x(\omega)|^2 \rangle &= x_0^2 \cdot 2 \operatorname{Re} \int_0^\infty dt \int_0^\infty e^{-\Gamma(2t+s)/2} e^{i(\omega-\omega_0)s} e^{-\nu s} ds \\ &= x_0^2 \cdot 2 \operatorname{Re} \int_0^\infty e^{-\Gamma t} dt \int_0^\infty \exp \left\{ -\left[ \frac{\Gamma}{2} + \nu - i(\omega - \omega_0) \right] s \right\} ds \\ &= \frac{x_0^2}{\Gamma} \left[ \frac{1}{\Gamma/2 + \nu - i(\omega - \omega_0)} + \frac{1}{\Gamma/2 + \nu + i(\omega - \omega_0)} \right] \\ &= \frac{x_0^2}{\Gamma} \left[ \frac{\Gamma + 2\nu}{(\omega - \omega_0)^2 + (\Gamma/2 + \nu)^2} \right] \end{aligned} \quad (7)$$

The energy spectrum is proportional to (7). The constant is fixed by requiring the total area to equal the radiated energy  $I_0$ . With

$$\int_{-\infty}^\infty \left[ \frac{\Gamma/2 + \nu}{(\omega - \omega_0)^2 + (\Gamma/2 + \nu)^2} \right] d\omega = \pi \quad (8)$$

we have

$$\frac{dI(\omega)}{d\omega} = \frac{I_0}{2\pi} \left[ \frac{\Gamma + 2\nu}{(\omega - \omega_0)^2 + (\Gamma/2 + \nu)^2} \right] \quad (9)$$

2. For the sodium D line,  $\lambda = 5893 \text{ \AA}$ ,  $T = 500 \text{ K}$ ,  $M = 23 \text{ amu} = 3.82 \times 10^{-23} \text{ g}$ , natural width  $\Delta\lambda_{\text{nat}} = 1.2 \times 10^{-4} \text{ \AA}$ . An atom with velocity component  $v$  along the line of sight emits at a shifted wavelength due to Doppler effect,

$$\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \quad (10)$$

At temperature  $T$  the line-of-sight velocity has the Maxwell-Boltzmann distribution

$$P(v) \propto \exp\left(-\frac{Mv^2}{2kT}\right) \quad (11)$$

so the velocities are spread over a Gaussian. Substituting  $v = c\Delta\lambda/\lambda$  turns the spread of velocities into a spread of wavelengths,

$$P(\Delta\lambda) \propto \exp\left(-\frac{Mc^2}{2kT} \frac{\Delta\lambda^2}{\lambda^2}\right) = \exp\left(-\frac{\Delta\lambda^2}{2a^2}\right) \quad a = \lambda \sqrt{\frac{kT}{Mc^2}} \quad (12)$$

so the line is a Gaussian of width  $a$  in wavelength. It falls to half its peak at  $\Delta\lambda = \pm a\sqrt{2\ln 2}$ , so the width between those two points is

$$\Delta\lambda_D = 2a\sqrt{2\ln 2} = \lambda \sqrt{\frac{8kT \ln 2}{Mc^2}} = 5893 \text{ \AA} \times 3.34 \times 10^{-6} \approx 0.020 \text{ \AA} \quad (13)$$

The Doppler width exceeds the natural width by a factor

$$\frac{\Delta\lambda_D}{\Delta\lambda_{\text{nat}}} \approx \frac{0.020}{1.2 \times 10^{-4}} \approx 1.6 \times 10^2 \quad (14)$$

Doppler broadening dominates at these conditions.

The sodium vapor is treated as an ideal gas, therefore

$$PV = NkT \quad \Rightarrow \quad n = \frac{N}{V} = \frac{P}{kT} \quad (15)$$

A collision depends on the motion of one atom relative to another, so the rate is set by the relative velocity  $\mathbf{v}_{\text{rel}} = \mathbf{v}_1 - \mathbf{v}_2$ . The single-atom speed follows the Maxwell-Boltzmann distribution

$$f(v) = 4\pi \left(\frac{M}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{Mv^2}{2kT}\right) \quad (16)$$

the mean of which is

$$\bar{v} = \int_0^\infty v f(v) dv = \sqrt{\frac{8kT}{\pi M}} \quad (17)$$

The relative velocity of two atoms of mass  $M$  follows the same distribution with the reduced mass  $\mu = M/2$ , so

$$\bar{v}_{\text{rel}} = \sqrt{\frac{8kT}{\pi\mu}} = \sqrt{\frac{16kT}{\pi M}} \quad (18)$$

The collision rate is thus

$$\nu = n\sigma\bar{v}_{\text{rel}} = \frac{\sigma P}{kT} \sqrt{\frac{16kT}{\pi M}} \quad (19)$$

From part (a) the collision contribution to the full width in angular frequency is  $\Delta\omega = 2\nu$ , which gives

$$\omega = \frac{2\pi c}{\lambda} \quad \Rightarrow \quad 2\nu = \Delta\omega = \left|\frac{d\omega}{d\lambda}\right| \Delta\lambda_c = \frac{2\pi c}{\lambda^2} \Delta\lambda_c \quad \Rightarrow \quad \Delta\lambda_c = \frac{\lambda^2 \nu}{\pi c} = \frac{\sigma \lambda^2 P}{\pi c kT} \sqrt{\frac{16kT}{\pi M}} \quad (20)$$

which allows us to find the pressure  $P$  needed to produce a given width  $\Delta\lambda$ :

$$P = \frac{\pi c kT}{\sigma \lambda^2} \sqrt{\frac{\pi M}{16kT}} \Delta\lambda \quad (21)$$

Evaluating with numerical values gives:

$$\Delta\lambda = \Delta\lambda_{\text{nat}} = 1.2 \times 10^{-4} \text{ \AA} \quad \Rightarrow \quad P \approx 2.3 \times 10^5 \text{ dyne/cm}^2 \quad (22)$$

$$\Delta\lambda = \Delta\lambda_D = 0.020 \text{ \AA} \quad \Rightarrow \quad P \approx 3.9 \times 10^7 \text{ dyne/cm}^2 \quad (23)$$